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Welfare programs and labor supply in developing countries: Experimental evidence from Latin America

Online appendix

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Abstract: This study looks at the effect of welfare programs on work incentives and the adult labor supply in developing countries. The analysis builds on the experimental evaluations of three programs implemented in rural areas: Mexico's PROGRESA, Nicaragua's *Red de Protección Social* ("Social Protection Network") (RPS) and Honduras' *Programa de Asignación Familiar* ("Family Allowance Program") (PRAF). The impact of welfare on labor supply has been widely studied in developed countries, where most recent initiatives attempt to mitigate negative effects on work incentives. The programs under study are conditional cash transfer (CCT) programs, which combine monetary benefits with incentives for curbing child labor and fostering the accumulation of human capital. Unlike their counterparts in developed economies, however, they do not account for potential impacts on the adult labor supply, and there is little systematic evidence concerning this aspect, despite a wealth of empirical studies on the intended outcomes. Comparable results for the three countries indicate that the effects that the programs have had on the labor supply of participating adults have been mostly negative but are nonetheless small and not statistically significant. Even though they have provided considerable transfers, the programs have not reduced the labor supply substantially in the short term. However, the evidence does point to the presence of other effects on labor markets. In the case of PROGRESA, there is a small positive effect on the number of hours worked by female beneficiaries and, after two years of implementation, a sizeable increase in wages among male beneficiaries and a resulting increase in household labor income. Moreover, PROGRESA seems to have reduced female labor-force participation in ineligible households in the localities randomly assigned to the program, while increasing ineligible males' participation in agricultural work. This mechanism is related to recent findings about the indirect impact of CCTs on ineligible households and implies that large-scale interventions may have broader equilibrium effects.

Keywords: welfare programs, income support, labor supply, work incentives, conditional cash transfers, randomized control trials, developing countries.

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Appendix: A Brief Description of the Programs

Mexico: PROGRESA Program

In 1997, Mexico began implementing the first phase of the PROGRESA (later renamed the *Oportunidades*) conditional cash transfer program in rural areas. The program had a multisector focus, with intended impacts on education, health and nutrition, in addition to the potential for poverty alleviation as a result of the cash transfer itself.

The initial deployment of the program was designed to facilitate the evaluation of its impact. The program was geographically targeted by locality, based on a poverty index. From an initial group of 506 localities that were selected for the first round, 320 were randomly selected to participate in the PROGRESA program (i.e., qualifying households in those localities would be eligible to participate); the program was not deployed in the remaining 186 localities. Households in the latter localities were still subject to the data collection process and thus constituted the control group for the program's evaluation.[1] Although the program was later expanded to cover additional areas, this study has focused on the initial deployment stage (1997-1999).

The data employed in this study have been drawn from the PROGRESA Evaluation Survey (ENCEL-*Encuesta de Evaluación de los Hogares*). The estimates given here are based on the initial baseline survey and three follow-up rounds that were conducted on a biannual basis following the program's introduction. The surveys collected socio-demographic and labor-market information at the household and individual levels for both the treatment and control localities.

Honduras: PRAF Program

The *Programa de Asignación Familiar* ("Family Allowance Program") (PRAF) was created by the Government of Honduras in the early 1990s as a compensatory mechanism to mitigate the impact of macroeconomic adjustments on the poor and to alleviate structural poverty. Its coverage was expanded a number of times until it ultimately reached a target population of 173,000 households with children from 0 to 14 years of age in 2008, and it now constitutes one of the largest

[1] The evaluation followed a phased-in process: PROGRESA was deployed in the control localities when the program's coverage was expanded in 2000.

welfare programs in the country. The objective of the program is to encourage poor households to invest in human capital –primarily education and health– through conditional cash transfers.

This study concentrates on the second phase of the program (PRAF II), which entailed a reorganization of the original intervention that had been planned in the late 1990s (Glewwe and Olinto, 2004). The PRAF II phase was implemented in 2000 and was geographically targeted at the municipality level in the poorest region of the country (IFPRI, 2000). It was deployed in a set of 50 randomly selected municipalities out of a total of 70, with the other 20 municipalities forming the control group. While the experimental design of PROGRESA represented an attempt to facilitate an evaluation of the overall impact of the program, the PRAF design was more ambitious. The original intention was to permit an evaluation of different types of interventions, and three sub-groups were therefore created within the treatment group: (i) municipalities scheduled to receive a demand-side intervention (cash subsidies), (ii) those scheduled to receive a supply-side intervention (i.e., construction of schools and health centers), and (iii) a group that would receive both. The empirical results presented below are based on the control group and the municipalities in the first treatment sub-group, since the supply-side interventions were never implemented (Glewwe and Olinto, 2004), and there are very few municipalities in the combined-intervention group.[2] A total of 40 municipalities were used for the estimation, with 20 of those municipalities being PRAF-eligible households and the other 20 being ones in which the program was not deployed.

The data used in this study were drawn from a baseline survey carried out in the second half of 2000 and a follow-up survey conducted in 2002, with a reasonably low sample attrition rate of approximately 8 percent. In contrast to the case of PROGRESA, where all households in treatment and control localities were interviewed, the PRAF surveys covered a sample of households. The corresponding sampling weights are used when possible in the empirical work outlined below.

[2] For more detailed information, see Glewwe and Olinto (2004).

Nicaragua: RPS Program

The *Red de Protección Social* ("Social Protection Network") (RPS) conditional cash transfer program was launched in 2000. The first phase consisted of a three-year pilot in two rural areas of the central region of Nicaragua (Madriz and Matagalpa), where poverty rates are above the national average. The program was broadly modeled after PROGRESA, and its main objective was to improve households' human capital through conditional cash transfers.

The 42 localities ("*comarcas*") with the lowest ratings on a multidimensional marginality index within the intervention area were selected for the pilot. Half of those localities were randomly assigned to the treatment group and the other half to the control group (Maluccio and Flores, 2005). The program was originally scheduled to be deployed in the control group localities after a year, but, due to a series of delays, they were not brought into the program until two years later.

The data used in this document were drawn from the initial baseline survey, which was carried out in the third quarter of 2000, and the first and second follow-ups, which were conducted in October 2001 and October 2002, respectively. The sample attrition rate was approximately 7 percent. As with the PRAF evaluation data, the survey covered a sample of the targeted population, and sampling weights have been used when possible.

Appendix: Analysis of the Random Assignment Process

Tables A1-A3 present the results of a probit regression of the probability of being selected into the treatment sample for each program as a function of observable household and individual characteristics. Since the focus of this study is on estimating differentiated treatment effects across population subgroups, estimates are presented for the entire adult population aged 15-65, as well as separately by gender, with standard errors being clustered at the locality level.[3] The results reveal some significant differences between treatment and control groups under all three programs.

The results for PRAF indicate that the situation in the treatment and control localities differed in two important ways. Households in the treatment sample had a significantly higher proportion of children attending school and a significantly lower proportion of children who were employed. There do not seem to be any significant differences by gender.

In localities selected for PROGRESA deployment, there was a significantly higher proportion of individuals who were employed, and this effect was accounted for mainly by men. The treatment and control samples for RPS, according to the probit regression, seem to be more balanced. None of the variables that were included in the analysis appears to be significantly associated with the probability of participation in the program.

[3] A number of other models were estimated using other disaggregated characteristics. However, the results were qualitatively similar. Some of the differences disappear once the probit regression accounts for clustering at the locality level.

Table A1-Probit Estimates for Treatment (Marginal Effects), Baseline Year: PRAF

Variable	All	Men	Women
Age	0.0006	0.0011	0.0005
	(0.001)	(0.001)	(0.001)
Gender (1= Male, 0= Female)	0.0302		
	(0.028)		
Number of children	0.0021	0.0001	0.0028
	(0.009)	(0.009)	(0.009)
Children employed	-0.0865	-0.0930	-0.0813
	(0.0270)***	(0.0297)***	(0.0284)***
Children attending school	0.1032	0.1249	0.0909
	(0.0375)***	(0.0371)***	(0.0400)**
Literacy (1= Yes, 0= No)	0.0019	0.0372	-0.0216
	(0.036)	(0.034)	(0.045)
Employed (1= Yes, 0= No)	-0.0109	-0.0450	0.0427
	(0.035)	(0.046)	(0.038)
Observations	6,897	2,868	4,029
LR Chi2	30.62	30.65	24.16

Source: Own calculations based on program evaluation survey.
Standard errors, clustered at the locality level, are shown in parentheses.
* Significant at 10%; ** significant at 5%; *** significant at 1%.

Table A2-Probit Estimates for Treatment (Marginal Effects), Baseline Year: PROGRESA

Variable	All	Men	Women
Age	-0.0001	0.0002	-0.0003
	(0.000)	(0.001)	(0.000)
Gender (1= Male, 0= Female)	-0.0155		
	(0.011)		
Number of children	0.0038	0.0050	0.0032
	(0.004)	(0.004)	(0.004)
Children employed	-0.0113	-0.0192	-0.0067
	(0.012)	(0.013)	(0.012)
Children attending school	0.0019	0.0039	-0.0001
	(0.015)	(0.016)	(0.015)
Literacy (1= Yes, 0= No)	-0.0178	-0.0089	-0.0266
	(0.026)	(0.029)	(0.025)
Employed (1= Yes, 0= No)	0.0311	0.0443	0.0160
	(0.0152)**	(0.0216)**	(0.016)
Observations	66,646	33,257	33,389
LR Chi2	9.63	8.71	2.48

Source: Own calculations based on program evaluation survey.
Standard errors, clustered at the locality level, are shown in parentheses.* Significant at 10%; ** significant at 5%; *** significant at 1%.

Table A3-Probit Estimates for Treatment (Marginal Effects), Baseline Year: RPS

Variable	All	Men	Women
Age	-0.0001	-0.0001	-0.0001
	(0.001)	(0.001)	(0.001)
Gender (1= Male, 0= Female)	0.0217		
	(0.028)		
Number of children	-0.0010	0.0008	-0.0029
	(0.012)	(0.012)	(0.012)
Children employed	-0.0509	-0.0623	-0.0390
	(0.038)	(0.041)	(0.038)
Children attending school	0.0349	0.0322	0.0388
	(0.044)	(0.044)	(0.046)
Literacy (1= Yes, 0= No)	0.0152	0.0118	0.0191
	(0.033)	(0.043)	(0.032)
Employed (1= Yes, 0= No)	-0.0188	-0.0292	0.0068
	(0.035)	(0.046)	(0.049)
Observations	4,674	2,290	2,384
LR Chi2	3.96	3.8	2.31

Source: Own calculations based on program evaluation survey.
Standard errors, clustered at the locality level, are shown in parentheses.* Significant at 10%; ** significant at 5%; *** significant at 1%.

Appendix: Analysis of Hours of Work and Household Composition: RPS

Table A4
Program Effects on Hours Worked by Adults in the Household and on the Number of Adults in the Household: RPS

	Dependent variable: Number of adults in the household					
Baseline: Aug-Sept. 2000	ITT		ITT Males		ITT Females	
	OLS	FE	OLS	FE	OLS	FE
t=1 (Oct. 2001)	**-0.051**	**-0.008**	**-0.027**	**0.012**	**-0.295**	**-0.145**
	(0.045)	(0.033)	(0.044)	(0.032)	(0.137)**	(0.099)
t=2 (Oct. 2002)	**-0.063**	**-0.057**	**-0.050**	**-0.048**	**-0.190**	**-0.065**
	(0.055)	(0.039)	(0.056)	(0.039)	(0.178)	(0.172)
Observations	4,124	4,124	3,652	3,652	472	472
Groups		1,525		1,331		194
	Dependent variable: Total number of hours worked by adults in the household					
Baseline: Aug-Sept. 2000						
t=1 (Oct. 2001)	**-4.444**	**-5.174**	**-3.447**	**-4.356**	**-12.338**	**-10.627**
	(2.925)	(2.787)*	(2.889)	(2.861)	(8.137)	(8.151)
t=2 (Oct. 2002)	**-4.769**	**-6.380**	**-3.546**	**-5.211**	**-16.023**	**-14.832**
	(2.823)*	(2.904)**	(2.847)	(2.899)*	(6.565)**	(6.896)**
Observations	4,124	4,124	3,652	3,652	472	472
Groups		1,525		1,331		194

Source: Own calculations based on program evaluation survey.
Standard errors, clustered at the locality level, are shown in parentheses.* Significant at 10%; ** significant at 5%; *** significant at 1%.